BAFU–ecoinvent

Elementary Flow Mapping

Proxy Mapping Change Log

Author: Anish Koyamparambath
Date: 11 June 2026
Source files: flows.csv

This document records only those final mappings where the BAFU elementary flow was not represented by an exact ecoinvent elementary flow and a proxy or proxy-like substitution was therefore used. Exact name matches, exact synonym matches, and unresolved no-match rows are outside the scope of this log.

1. Selection Rule and Audit Framing

The main selection was made from Last adjustments.xlsx, sheet Corrected_Mapping. Rows were included when mapping_method was proxy, proxy_unit_conversion, or exact_proxy_category. The first two classes are direct proxy classes. The last class was included because it changes the BAFU category logic while keeping an actual ecoinvent elementary flow. Duplicate compartment rows were consolidated into one entry per BAFU flow, ecoinvent target, and mapping method, with all affected categories retained.

This change log follows the same documentation logic required for modified BAFU data: the altered data point, the nature of the change, and the rationale for each modification are made explicit. BAFU Terms of Use sections 3.2 and 3.3 also state that modification documentation should be provided in a standalone report accompanying the modified dataset.

2. Consolidated Proxy Index

The table below is an index. The detailed log in Section 3 gives the scientific and practical explanation for each mapping.

#	BAFU Flow	ecoinvent Proxy	Compartment/ Sub Compartment
1	Energy, waste heat, air	Heat, waste	natural resource / in air
2	4,4'-Biphenol	Phenol	air
3	Benzene, dichloro-	o-Dichlorobenzene	air / urban air close to ground
4	Butadiene, hexachloro-	Chlorinated solvents, unspecified	water / surface water; water
5	Butane, perfluorocyclo-, PFC-318	Perfluoropentane	air / urban air close to ground
6	Calcium chloride	Calcium	natural resource / in ground
7	Caprolactam	Acetamide	air
8	Chemically polluted water	COD, Chemical Oxygen Demand	water / surface water
9	Cresol	Phenol	water / surface water; water / ocean
10	Decane	Hydrocarbons, aliphatic, alkanes, unspecified	water / ocean; water / surface water; soil / industrial
11	Disodium acid methane arsenate	Arsenic ion	soil / agricultural
12	Energy, gross calorific value, in biomass, resource correction	Energy, gross calorific value, in biomass	natural resource / biotic; resources
13	Fluorine, 4.5% in apatite, 3% in crude ore	Fluorine	natural resource / in ground
14	Kerosene	Oils, unspecified	air / urban air close to ground
15	m-Cresol	Phenol	air
16	Methyl mercaptan	Butane-1-thiol	air
17	Monosodium acid methanearsonate	Arsenic ion	soil / agricultural
18	Occupation, traffic area	Occupation, traffic area, road network	resources
19	Occupation, urban	Occupation, urban, discontinuously built	resources
20	Octene (mixture of isomers)	Hydrocarbons, aliphatic, unsaturated	air

#	BAFU Flow	ecoinvent Proxy	Compartment/ Sub Compartment
21	p-Cresol	Phenol	air
22	Petrol	Oils, unspecified	water / surface water
23	Phosphorus, 18% in apatite, 4% in crude ore	Phosphorus	natural resource / in ground
24	Propane, 1,1,1,3,3-pentafluoro-, HFC-245fa	Pentafluoroethane	air
25	Propane, perfluorocyclo-	Hexafluoroethane	air / urban air close to ground
26	Propylene glycol methyl ether acetate	NMVOC, non-methane volatile organic compounds	air
27	Sulfide	Sulfur	soil / industrial
28	Thiazole, 2-(thiocyanatemethylthio)benzo-	Pesticides, unspecified	soil / agricultural
29	Water, well, RER	Water, well, in ground	natural resource / in water; natural resource / in ground
30	Energy, from coal	Coal, hard	natural resource / in ground
31	Energy, from coal, brown	Coal, brown	natural resource / in ground
32	Energy, from oil	Oil, crude	resources
33	Energy, from peat	Peat	natural resource / in ground
34	Energy, from uranium	Uranium	natural resource / in ground
35	Water, process, unspecified natural origin/kg	Water, unspecified natural origin	natural resource / land; natural resource / in ground; resources
36	Wood, unspecified, standing/kg	Wood, unspecified, standing	natural resource / land; resources

3. Detailed Proxy Change Log

1. Energy, waste heat, air → Heat, waste

BAFU source flow	Energy, waste heat, air
ecoinvent proxy flow	Heat, waste
Proxy class	Category proxy / reclassification
Affected BAFU category	natural resource / in air
Target compartment	water / unspecified
Target unit	MJ
Target CAS / formula	not available / not available
Unit / compartment / sub-compartment match	Unit: Yes | Compartment: No | Sub-compartment: No
Exchange count / dataset count	1 / 1
Review priority	Low
ecoinvent ID	9b83f2bb-7d61-4e55-b37a-443f164c5f30

Rationale: Waste heat exists as an actual ecoinvent flow; ecoinvent treats it as an emission to air, not a natural resource.

Limitation: Compartment and sub-compartment differ from the BAFU source category.

2. 4,4'-Biphenol → Phenol

BAFU source flow	4,4'-Biphenol
ecoinvent proxy flow	Phenol
Proxy class	Direct proxy
Affected BAFU category	air
Target compartment	air / unspecified
Target unit	kg
Target CAS / formula	000108-95-2 / C6H6O
Unit / compartment / sub-compartment match	Unit: Yes | Compartment: Yes | Sub-compartment: Yes
Exchange count / dataset count	7 / 7
Review priority	Low
ecoinvent ID	5896c761-4e02-4573-8747-c4bbf73162c0

Rationale: No actual ecoinvent flow for 4,4'-biphenol / 4,4'-dihydroxybiphenyl; Phenol is used as a low-confidence phenolic aromatic proxy. The selected target preserves the phenolic aromatic functional group, which is the dominant chemical similarity relevant for a screening-level environmental-flow proxy.

Limitation: Proxy is suitable for screening only; compartment and unit are retained.

3. Benzene, dichloro- → o-Dichlorobenzene

BAFU source flow	Benzene, dichloro-
ecoinvent proxy flow	o-Dichlorobenzene
Proxy class	Direct proxy
Affected BAFU category	air / urban air close to ground
Target compartment	air / urban air close to ground
Target unit	kg
Target CAS / formula	000095-50-1 / C6H4Cl2
Unit / compartment / sub-compartment match	Unit: Yes | Compartment: Yes | Sub-compartment: Yes
Exchange count / dataset count	82 / 82
Review priority	Medium
ecoinvent ID	9645e02f-855a-4b9f-8baf-f34a08fa80c4

Rationale: Unspecified dichlorobenzene isomer; ecoinvent contains o-Dichlorobenzene as an actual flow and this is used as an isomer proxy.

Limitation: Proxy is suitable for screening if the target compartment and unit are retained.

4. Butadiene, hexachloro- → Chlorinated solvents, unspecified

BAFU source flow	Butadiene, hexachloro-
ecoinvent proxy flow	Chlorinated solvents, unspecified
Proxy class	Direct proxy
Affected BAFU category	water / surface water; water
Target compartment	water / surface water; unspecified
Target unit	kg
Target CAS / formula	not available / not available
Unit / compartment / sub-compartment match	Unit: Yes | Compartment: Yes | Sub-compartment: Yes
Exchange count / dataset count	2; 7 / 2; 7
Review priority	Low
ecoinvent ID	0561e8f7-bb2b-45b7-b4ab-e750f6842981; f6f5933a-5119-41aa-a5ec-7919706c59f2

Rationale: No actual hexachlorobutadiene flow in the provided ecoinvent exchanges; mapped to actual chlorinated-solvents group flow in the same environmental compartment. This group-flow proxy preserves the broad substance class and environmental compartment when no substance-specific ecoinvent flow is available.

Limitation: Proxy is suitable for screening if the target compartment and unit are retained.

5. Butane, perfluorocyclo-, PFC-318 → Perfluoropentane

BAFU source flow	Butane, perfluorocyclo-, PFC-318
ecoinvent proxy flow	Perfluoropentane
Proxy class	Direct proxy
Affected BAFU category	air / urban air close to ground
Target compartment	air / unspecified
Target unit	kg
Target CAS / formula	000678-26-2 / C5F12
Unit / compartment / sub-compartment match	Unit: Yes | Compartment: Yes | Sub-compartment: No
Exchange count / dataset count	64 / 64
Review priority	Medium
ecoinvent ID	e2ccc98a-6d4e-443b-b11f-d16f32782833

Rationale: No actual perfluorocyclobutane / PFC-318 flow in the provided ecoinvent exchanges; Perfluoropentane is the closest actual perfluorocarbon proxy present. The target retains the fluorinated-organic character, but its atmospheric behaviour or characterization factor may differ from the original flow.

Limitation: Sub-compartment differs or is unavailable.

6. Calcium chloride → Calcium

BAFU source flow	Calcium chloride
ecoinvent proxy flow	Calcium
Proxy class	Direct proxy
Affected BAFU category	natural resource / in ground
Target compartment	natural resource / in ground
Target unit	kg
Target CAS / formula	7440-70-2 / Ca
Unit / compartment / sub-compartment match	Unit: Yes | Compartment: Yes | Sub-compartment: Yes
Exchange count / dataset count	1 / 1
Review priority	Low
ecoinvent ID	c8fc4197-7410-42f2-aeb4-c08c6a693992

Rationale: No actual calcium chloride natural-resource flow exists; mapped to actual Calcium natural resource as elemental resource proxy.

Limitation: Mass / stoichiometric conversion should be checked before numerical use.

7. Caprolactam → Acetamide

BAFU source flow	Caprolactam
ecoinvent proxy flow	Acetamide
Proxy class	Direct proxy
Affected BAFU category	air
Target compartment	air / non-urban air or from high stacks
Target unit	kg
Target CAS / formula	000060-35-5 / C2H5NO
Unit / compartment / sub-compartment match	Unit: Yes | Compartment: Yes | Sub-compartment: Yes
Exchange count / dataset count	1 / 1
Review priority	Low
ecoinvent ID	c3de45a3-e6fd-494e-aa85-d352b478e87b

Rationale: No actual caprolactam flow exists; Acetamide is an actual amide flow and is used as a low-confidence functional-group proxy.

Limitation: Proxy is suitable for screening if the target compartment and unit are retained.

8. Chemically polluted water → COD, Chemical Oxygen Demand

BAFU source flow	Chemically polluted water
ecoinvent proxy flow	COD, Chemical Oxygen Demand
Proxy class	Direct proxy
Affected BAFU category	water / surface water
Target compartment	water / surface water
Target unit	kg
Target CAS / formula	not available / not available
Unit / compartment / sub-compartment match	Unit: Yes | Compartment: Yes | Sub-compartment: Yes
Exchange count / dataset count	84 / 84
Review priority	Medium
ecoinvent ID	fc0b5c85-3b49-42c2-a3fd-db7e57b696e3

Rationale: No actual chemically polluted water flow exists; COD is an actual water-pollution indicator used as a proxy. It is suitable only where the BAFU flow represents chemically polluted water as an aggregate organic or oxidizable load.

Limitation: Proxy is suitable for screening if the target compartment and unit are retained.

9. Cresol → Phenol

BAFU source flow	Cresol

ecoinvent proxy flow	Phenol
Proxy class	Direct proxy
Affected BAFU category	water / surface water; water / ocean
Target compartment	water / ocean; surface water
Target unit	kg
Target CAS / formula	000108-95-2 / C6H6O
Unit / compartment / sub-compartment match	Unit: Yes | Compartment: Yes | Sub-compartment: Yes
Exchange count / dataset count	1 / 1
Review priority	Low
ecoinvent ID	a3436836-e2fe-4c30-b9a7-0098b489374f; 64836dd0-8b0f-41ba-a7d3-be54358fc4b3

Rationale: No actual cresol flow exists; Phenol is used as a phenolic aromatic proxy, preserving the dominant functional group relevant for screening-level mapping.

Limitation: Proxy is suitable for screening if the target compartment and unit are retained.

10. Decane → Hydrocarbons, aliphatic, alkanes, unspecified

BAFU source flow	Decane
ecoinvent proxy flow	Hydrocarbons, aliphatic, alkanes, unspecified
Proxy class	Direct proxy
Affected BAFU category	water / ocean; water / surface water; soil / industrial
Target compartment	air; water / unspecified; ocean; surface water
Target unit	kg
Target CAS / formula	not available / not available
Unit / compartment / sub-compartment match	Unit: Yes | Compartment: No | Sub-compartment: No
Exchange count / dataset count	1 / 1
Review priority	Low
ecoinvent ID	2451f9df-0596-49ec-8380-08e4ad75a153; faf05b06-da29-4af5-aa37-cfa4b802a297; b902d333-ded6-4c87-be52-8fc03fe6dfd4

Rationale: No actual decane flow exists; mapped to actual aliphatic alkane group flow in the same compartment. This group-flow proxy preserves the broad substance class when no substance-specific ecoinvent flow is available.

Limitation: Compartment and sub-compartment differ from the BAFU source category.

11. Disodium acid methane arsenate → Arsenic ion

BAFU source flow	Disodium acid methane arsenate
ecoinvent proxy flow	Arsenic ion
Proxy class	Direct proxy
Affected BAFU category	soil / agricultural
Target compartment	soil / agricultural
Target unit	kg
Target CAS / formula	not available / not available
Unit / compartment / sub-compartment match	Unit: Yes | Compartment: Yes | Sub-compartment: Yes
Exchange count / dataset count	2 / 2
Review priority	Low
ecoinvent ID	4673a799-36a5-4359-ac6a-fa933281bc93

Rationale: No actual DSMA / disodium methanearsonate flow exists; mapped to actual Arsenic ion in the same compartment as an elemental toxicity proxy. Compound-to-element conversion not applied. The elemental arsenic toxicity driver is more defensible than selecting an unrelated organic arsenic compound absent from the ecoinvent exchange list.

Limitation: Mass / stoichiometric conversion should be checked before numerical use.

12. Energy, gross calorific value, in biomass, resource correction → Energy, gross calorific value, in biomass

BAFU source flow	Energy, gross calorific value, in biomass, resource correction
ecoinvent proxy flow	Energy, gross calorific value, in biomass
Proxy class	Direct proxy
Affected BAFU category	natural resource / biotic; resources
Target compartment	natural resource / biotic
Target unit	MJ
Target CAS / formula	not available / not available
Unit / compartment / sub-compartment match	Unit: Yes | Compartment: Yes | Sub-compartment: Yes
Exchange count / dataset count	21 / 21
Review priority	Special review
ecoinvent ID	01c12fca-ad8b-4902-8b48-2d5afe3d3a0f

Rationale: No separate ecoinvent resource-correction flow in the elementary exchange list; mapped to actual biomass calorific-value flow as closest resource proxy. The target retains the same primary energy-resource family.

Limitation: Proxy is suitable for screening if the target compartment and unit are retained.

13. Fluorine, 4.5% in apatite, 3% in crude ore → Fluorine

BAFU source flow	Fluorine, 4.5% in apatite, 3% in crude ore
ecoinvent proxy flow	Fluorine
Proxy class	Direct proxy
Affected BAFU category	natural resource / in ground
Target compartment	natural resource / in ground
Target unit	kg
Target CAS / formula	007782-41-4 / F2
Unit / compartment / sub-compartment match	Unit: Yes | Compartment: Yes | Sub-compartment: Yes
Exchange count / dataset count	88 / 88
Review priority	Medium
ecoinvent ID	3048af84-1d72-5e3f-a739-b2d7fa7d4773

Rationale: No apatite-specific fluorine resource flow exists; mapped to actual Fluorine natural resource in ground. The target retains the fluorine-resource character, but its characterization factor may differ from the original flow.

Limitation: Proxy is suitable for screening if the target compartment and unit are retained.

14. Kerosene → Oils, unspecified

BAFU source flow	Kerosene
ecoinvent proxy flow	Oils, unspecified
Proxy class	Direct proxy
Affected BAFU category	air / urban air close to ground
Target compartment	water / unspecified
Target unit	kg
Target CAS / formula	not available / not available
Unit / compartment / sub-compartment match	Unit: Yes | Compartment: No | Sub-compartment: No
Exchange count / dataset count	64 / 64
Review priority	Medium
ecoinvent ID	6da0e891-3648-43a5-b0cf-f1872c57db8d

Rationale: No actual kerosene elementary flow exists; for environmental release, mapped to actual Oils, unspecified as a group-flow proxy. The proxy preserves the broad substance class and environmental compartment when no substance-specific ecoinvent flow is available.

Limitation: Compartment and sub-compartment differ from the BAFU source category.

15. m-Cresol → Phenol

BAFU source flow	m-Cresol
ecoinvent proxy flow	Phenol
Proxy class	Direct proxy
Affected BAFU category	air
Target compartment	air / unspecified
Target unit	kg
Target CAS / formula	000108-95-2 / C6H6O
Unit / compartment / sub-compartment match	Unit: Yes | Compartment: Yes | Sub-compartment: Yes
Exchange count / dataset count	1 / 1
Review priority	Low
ecoinvent ID	5896c761-4e02-4573-8747-c4bbf73162c0

Rationale: No actual m-cresol flow exists; Phenol is used as a phenolic aromatic proxy, preserving the dominant functional group relevant for screening-level mapping.

Limitation: Proxy is suitable for screening if the target compartment and unit are retained.

16. Methyl mercaptan → Butane-1-thiol

BAFU source flow	Methyl mercaptan
ecoinvent proxy flow	Butane-1-thiol
Proxy class	Direct proxy
Affected BAFU category	air
Target compartment	air / unspecified
Target unit	kg
Target CAS / formula	000109-79-5 / C4H10S
Unit / compartment / sub-compartment match	Unit: Yes | Compartment: Yes | Sub-compartment: Yes
Exchange count / dataset count	1 / 1
Review priority	Low
ecoinvent ID	82adbec8-225f-5292-8962-2875cee71298

Rationale: No actual methyl mercaptan / methanethiol flow exists; Butane-1-thiol is the closest actual thiol / mercaptan proxy in ecoinvent. Confidence is low.

Limitation: Low-confidence chemical proxy.

17. Monosodium acid methanearsonate → Arsenic ion

BAFU source flow	Monosodium acid methanearsonate
ecoinvent proxy flow	Arsenic ion
Proxy class	Direct proxy
Affected BAFU category	soil / agricultural
Target compartment	soil / agricultural
Target unit	kg
Target CAS / formula	not available / not available
Unit / compartment / sub-compartment match	Unit: Yes | Compartment: Yes | Sub-compartment: Yes
Exchange count / dataset count	4 / 4
Review priority	Low
ecoinvent ID	4673a799-36a5-4359-ac6a-fa933281bc93

Rationale: No actual MSMA / monosodium methanearsonate flow exists; mapped to actual Arsenic ion in the same compartment as an elemental arsenic toxicity proxy. Compound-to-element conversion not applied.

Limitation: Mass / stoichiometric conversion should be checked before numerical use.

18. Occupation, traffic area → Occupation, traffic area, road network

BAFU source flow	Occupation, traffic area
ecoinvent proxy flow	Occupation, traffic area, road network
Proxy class	Direct proxy
Affected BAFU category	resources
Target compartment	natural resource / land
Target unit	$m^2 \cdot year$
Target CAS / formula	not available / not available
Unit / compartment / sub-compartment match	Unit: Yes | Compartment: Yes | Sub-compartment: Yes
Exchange count / dataset count	10 / 10
Review priority	Low
ecoinvent ID	26efe47c-92a5-4dea-b4d0-eac13e418a58

Rationale: No generic traffic-area occupation flow exists; mapped to actual road-network traffic-area occupation as the closest specific land-use proxy.

Limitation: Proxy is suitable for screening if the target compartment and unit are retained.

19. Occupation, urban → Occupation, urban, discontinuously built

BAFU source flow	Occupation, urban
ecoinvent proxy flow	Occupation, urban, discontinuously built
Proxy class	Direct proxy
Affected BAFU category	resources
Target compartment	natural resource / land
Target unit	m^2·year
Target CAS / formula	not available / not available
Unit / compartment / sub-compartment match	Unit: Yes | Compartment: Yes | Sub-compartment: Yes
Exchange count / dataset count	17 / 17
Review priority	Low
ecoinvent ID	56ec994a-eb96-42e8-93eb-4970e30e6362

Rationale: No generic urban occupation flow exists; mapped to actual discontinuously built urban occupation as a broadly representative proxy.

Limitation: Proxy is suitable for screening if the target compartment and unit are retained.

20. Octene (mixture of isomers) → Hydrocarbons, aliphatic, unsaturated

BAFU source flow	Octene (mixture of isomers)
ecoinvent proxy flow	Hydrocarbons, aliphatic, unsaturated
Proxy class	Direct proxy
Affected BAFU category	air
Target compartment	air / unspecified
Target unit	kg
Target CAS / formula	not available / not available
Unit / compartment / sub-compartment match	Unit: Yes | Compartment: Yes | Sub-compartment: Yes
Exchange count / dataset count	2 / 2
Review priority	Low
ecoinvent ID	4cff006f-c6df-42a8-9dee-cc064f14326e

Rationale: No actual octene isomer-mixture flow exists; mapped to actual aliphatic unsaturated hydrocarbon group. This group-flow proxy preserves the broad substance class and environmental compartment.

Limitation: Proxy is suitable for screening if the target compartment and unit are retained.

21. p-Cresol → Phenol

BAFU source flow	p-Cresol
ecoinvent proxy flow	Phenol
Proxy class	Direct proxy
Affected BAFU category	air
Target compartment	air / unspecified
Target unit	kg
Target CAS / formula	000108-95-2 / C6H6O
Unit / compartment / sub-compartment match	Unit: Yes | Compartment: Yes | Sub-compartment: Yes
Exchange count / dataset count	1 / 1
Review priority	Low
ecoinvent ID	5896c761-4e02-4573-8747-c4bbf73162c0

Rationale: No actual p-cresol flow exists; Phenol is used as a phenolic aromatic proxy, preserving the dominant functional group relevant for screening-level mapping.

Limitation: Proxy is suitable for screening if the target compartment and unit are retained.

22. Petrol → Oils, unspecified

BAFU source flow	Petrol
ecoinvent proxy flow	Oils, unspecified
Proxy class	Direct proxy
Affected BAFU category	water / surface water
Target compartment	water / surface water
Target unit	kg
Target CAS / formula	not available / not available
Unit / compartment / sub-compartment match	Unit: Yes | Compartment: Yes | Sub-compartment: Yes
Exchange count / dataset count	64 / 64
Review priority	Medium
ecoinvent ID	82c27b2a-69a8-4b1e-ad4e-4de9475630c6

Rationale: No actual petrol / gasoline elementary flow exists; mapped to actual Oils, unspecified in the same water compartment as a group-flow proxy.

Limitation: Proxy is suitable for screening if the target compartment and unit are retained.

23. Phosphorus, 18% in apatite, 4% in crude ore → Phosphorus

BAFU source flow	Phosphorus, 18% in apatite, 4% in crude ore
ecoinvent proxy flow	Phosphorus
Proxy class	Direct proxy
Affected BAFU category	natural resource / in ground
Target compartment	natural resource / in ground
Target unit	kg
Target CAS / formula	7723-14-0 / P
Unit / compartment / sub-compartment match	Unit: Yes | Compartment: Yes | Sub-compartment: Yes
Exchange count / dataset count	88 / 88
Review priority	Medium
ecoinvent ID	483ae3c5-4eb0-46e4-b811-a72ad391716b

Rationale: Corrected target typo ("Phosphorous") to actual ecoinvent Phosphorus natural resource in ground.

Limitation: Proxy is suitable for screening if the target compartment and unit are retained.

24. Propane, 1,1,1,3,3-pentafluoro-, HFC-245fa → Pentafluoroethane

BAFU source flow	Propane, 1,1,1,3,3-pentafluoro-, HFC-245fa
ecoinvent proxy flow	Pentafluoroethane
Proxy class	Direct proxy
Affected BAFU category	air
Target compartment	air / unspecified
Target unit	kg
Target CAS / formula	000354-33-6 / C2HF5
Unit / compartment / sub-compartment match	Unit: Yes | Compartment: Yes | Sub-compartment: Yes
Exchange count / dataset count	1 / 1
Review priority	Low
ecoinvent ID	281e1011-f121-4d5b-ad4d-22d562b2c2cd

Rationale: No actual HFC-245fa / pentafluoropropane flow exists; Pentafluoroethane / HFC-125 is the closest actual HFC proxy present. The target retains the fluorinated-organic character, but its atmospheric behaviour or characterization factor may differ.

Limitation: Proxy is suitable for screening if the target compartment and unit are retained.

25. Propane, perfluorocyclo- → Hexafluoroethane

BAFU source flow	Propane, perfluorocyclo-
ecoinvent proxy flow	Hexafluoroethane
Proxy class	Direct proxy
Affected BAFU category	air / urban air close to ground
Target compartment	air / urban air close to ground
Target unit	kg
Target CAS / formula	000076-16-4 / C2F6
Unit / compartment / sub-compartment match	Unit: Yes | Compartment: Yes | Sub-compartment: Yes
Exchange count / dataset count	64 / 64
Review priority	Medium
ecoinvent ID	df5dd437-2e12-4af6-8f7a-9c8224857dc5

Rationale: No actual perfluorocyclopropane flow exists; Hexafluoroethane is the closest actual fluorocarbon proxy with a C-F only formula present in multiple air sub-compartments.

Limitation: Proxy is suitable for screening if the target compartment and unit are retained.

26. Propylene glycol methyl ether acetate → NMVOC, non-methane volatile organic compounds

BAFU source flow	Propylene glycol methyl ether acetate
ecoinvent proxy flow	NMVOC, non-methane volatile organic compounds
Proxy class	Direct proxy
Affected BAFU category	air
Target compartment	air / unspecified
Target unit	kg
Target CAS / formula	not available / not available
Unit / compartment / sub-compartment match	Unit: Yes | Compartment: Yes | Sub-compartment: Yes
Exchange count / dataset count	1 / 1
Review priority	Low
ecoinvent ID	d3260d0e-8203-4cbb-a45a-6a13131a5108

Rationale: No actual PGMEA flow exists; mapped to actual NMVOC group flow for air emission as a group-flow proxy.

Limitation: Proxy is suitable for screening if the target compartment and unit are retained.

27. Sulfide → Sulfur

BAFU source flow	Sulfide
ecoinvent proxy flow	Sulfur
Proxy class	Direct proxy
Affected BAFU category	soil / industrial
Target compartment	soil / industrial
Target unit	kg
Target CAS / formula	007704-34-9 / S
Unit / compartment / sub-compartment match	Unit: Yes | Compartment: Yes | Sub-compartment: Yes
Exchange count / dataset count	1 / 1
Review priority	Low
ecoinvent ID	16f40b1c-f7c8-4100-9916-81a260493951

Rationale: No actual generic sulfide flow exists; mapped to elemental Sulfur in the same compartment as a conservative elemental proxy.

Limitation: Proxy is suitable for screening if the target compartment and unit are retained.

28. Thiazole, 2-(thiocyanatemethylthio)benzo- → Pesticides, unspecified

BAFU source flow	Thiazole, 2-(thiocyanatemethylthio)benzo-
ecoinvent proxy flow	Pesticides, unspecified
Proxy class	Direct proxy
Affected BAFU category	soil / agricultural
Target compartment	soil / agricultural
Target unit	kg
Target CAS / formula	not available / not available
Unit / compartment / sub-compartment match	Unit: Yes | Compartment: Yes | Sub-compartment: Yes
Exchange count / dataset count	1 / 1
Review priority	Low
ecoinvent ID	18fb2184-d78e-467f-9eac-116bd534739f

Rationale: No actual TCMTB flow exists; mapped to actual Pesticides, unspecified in agricultural soil as a functional use-class proxy.

Limitation: Proxy is suitable for screening if the target compartment and unit are retained.

29. Water, well, RER → Water, well, in ground

BAFU source flow	Water, well, RER
ecoinvent proxy flow	Water, well, in ground
Proxy class	Direct proxy
Affected BAFU category	natural resource / in water; natural resource / in ground
Target compartment	natural resource / in water
Target unit	m^3
Target CAS / formula	007732-18-5 / H2O
Unit / compartment / sub-compartment match	Unit: Yes | Compartment: Yes | Sub-compartment: No
Exchange count / dataset count	6 / 6
Review priority	Low
ecoinvent ID	67c40aae-d403-464d-9649-c12695e43ad8

Rationale: Actual ecoinvent flow is Water, well, in ground; the regional RER qualifier is unavailable in the elementary exchange list.

Limitation: Sub-compartment differs or is unavailable.

30. Energy, from coal → Coal, hard

BAFU source flow	Energy, from coal
ecoinvent proxy flow	Coal, hard
Proxy class	Proxy with unit conversion
Affected BAFU category	natural resource / in ground
Target compartment	natural resource / in ground
Target unit	kg
Target CAS / formula	not available / not available
Unit / compartment / sub-compartment match	Unit: No | Compartment: Yes | Sub-compartment: Yes
Exchange count / dataset count	14 / 14
Review priority	Low
ecoinvent ID	b6d0042d-0ef8-49ed-9162-a07ff1ccf750

Rationale: No MJ-based coal energy resource flow exists in ecoinvent; mapped to actual hard coal natural resource. This is not a direct mass-for-mass substitution; a heating-value conversion factor must be applied before calculations.

Limitation: Unit conversion (MJ → kg) required before numerical use.

31. Energy, from coal, brown → Coal, brown

BAFU source flow	Energy, from coal, brown
ecoinvent proxy flow	Coal, brown
Proxy class	Proxy with unit conversion
Affected BAFU category	natural resource / in ground
Target compartment	natural resource / in ground
Target unit	kg
Target CAS / formula	not available / not available
Unit / compartment / sub-compartment match	Unit: No | Compartment: Yes | Sub-compartment: Yes
Exchange count / dataset count	14 / 14
Review priority	Low
ecoinvent ID	024c9722-1e88-412b-8c4b-10c532be8dca

Rationale: Brown coal corresponds to lignite / brown coal; actual ecoinvent flow is mass-based, so an MJ-to-kg conversion is required before use.

Limitation: Unit conversion (MJ → kg) required before numerical use.

32. Energy, from oil → Oil, crude

BAFU source flow	Energy, from oil
ecoinvent proxy flow	Oil, crude
Proxy class	Proxy with unit conversion
Affected BAFU category	resources
Target compartment	natural resource / in ground
Target unit	kg
Target CAS / formula	not available / not available
Unit / compartment / sub-compartment match	Unit: No | Compartment: Yes | Sub-compartment: Yes
Exchange count / dataset count	2 / 2
Review priority	Low
ecoinvent ID	88d06db9-59a1-4719-9174-afeb1fa4026a

Rationale: Energy from oil mapped to actual crude oil natural resource. Unit conversion from MJ to kg is required before use.

Limitation: Unit conversion (MJ → kg) required before numerical use.

33. Energy, from peat → Peat

BAFU source flow	Energy, from peat
ecoinvent proxy flow	Peat
Proxy class	Proxy with unit conversion
Affected BAFU category	natural resource / in ground
Target compartment	natural resource / biotic
Target unit	kg
Target CAS / formula	not available / not available
Unit / compartment / sub-compartment match	Unit: No | Compartment: Yes | Sub-compartment: No
Exchange count / dataset count	1 / 1
Review priority	Low
ecoinvent ID	c5035ce2-5ee5-431f-a287-4b25da42be74

Rationale: Energy from peat mapped to actual peat resource; ecoinvent uses kg, so an MJ-to-kg conversion is required.

Limitation: Unit conversion (MJ → kg) required; sub-compartment differs or is unavailable.

34. Energy, from uranium → Uranium

BAFU source flow	Energy, from uranium
ecoinvent proxy flow	Uranium
Proxy class	Proxy with unit conversion
Affected BAFU category	natural resource / in ground
Target compartment	natural resource / in ground
Target unit	kg
Target CAS / formula	007440-61-1 / U
Unit / compartment / sub-compartment match	Unit: No | Compartment: Yes | Sub-compartment: Yes
Exchange count / dataset count	7 / 7
Review priority	Low
ecoinvent ID	2ba5e39b-adb6-4767-a51d-90c1cf32fe98

Rationale: Energy from uranium mapped to actual uranium natural resource. Unit conversion from MJ to kg uranium is required.

Limitation: Unit conversion (MJ → kg) required before numerical use.

35. Water, process, unspecified natural origin/kg → Water, unspecified natural origin

BAFU source flow	Water, process, unspecified natural origin/kg
ecoinvent proxy flow	Water, unspecified natural origin
Proxy class	Proxy with unit conversion
Affected BAFU category	natural resource / land; natural resource / in ground; resources
Target compartment	natural resource / in water
Target unit	m^3
Target CAS / formula	007732-18-5 / H2O
Unit / compartment / sub-compartment match	Unit: No | Compartment: Yes | Sub-compartment: No
Exchange count / dataset count	64 / 64
Review priority	Medium
ecoinvent ID	831f249e-53f2-49cf-a93c-7cee105f048e

Rationale: Actual ecoinvent water flow exists only in m^3; mapping retained with a unit conversion note. The water-resource identity is preserved, but the source mass unit must be converted to a volume unit using an explicit density assumption.

Limitation: Unit conversion (kg → m^3) required; sub-compartment differs or is unavailable.

36. Wood, unspecified, standing/kg → Wood, unspecified, standing

BAFU source flow	Wood, unspecified, standing/kg
ecoinvent proxy flow	Wood, unspecified, standing
Proxy class	Proxy with unit conversion
Affected BAFU category	natural resource / land; resources
Target compartment	natural resource / biotic
Target unit	m^3
Target CAS / formula	not available / not available
Unit / compartment / sub-compartment match	Unit: No | Compartment: Yes | Sub-compartment: No
Exchange count / dataset count	1 / 1
Review priority	Low
ecoinvent ID	23e83c1f-07c9-4b5f-a898-0f4f09a6691f

Rationale: Actual ecoinvent wood standing resource exists only in m^3; mapping retained with unit conversion note. The standing-wood resource identity is preserved, but the source mass unit and the ecoinvent volume unit require an explicit conversion assumption.

Limitation: Unit conversion (kg → m^3) required; sub-compartment differs or is unavailable.

4. Interpretation Notes

A proxy mapping should not be read as a claim that the BAFU flow and the ecoinvent target are chemically identical. It is a documented replacement used because the actual ecoinvent elementary exchange list does not contain the exact BAFU flow.

Where the proxy uses a group flow, the mapping is suitable mainly for screening or inventory completeness. Where the proxy uses an elemental flow, the elemental driver is retained, but compound-to-element mass conversion may still be required. Where a unit mismatch exists, the mapped flow should not be used numerically until an explicit conversion factor is applied.

For chemical substances with strong fate, toxicity, or climate-forcing differences, the mapped proxy should be treated as a temporary bridge until a better ecoinvent elementary flow or a new custom elementary flow can be justified. The low-confidence rows should be reviewed first if they have high contribution in LCIA results.